

CS5095

Types and Programming Languages

Lecture 2: Judgments, Inference Rules and Rule Induction

Announcements

- Discord
- Homework 1
- Office hours

We state things with *judgments*

logical statements, properties of programs, math equations, ...

- Judgment: **statement** with variables (*metavariables*) standing in for other things/expressions

Important: the parts in blue don't change

- Examples:

- $X + Y = Z$
 - $1 + 2 = 3$
- **e is a natural number**
 - 1 is a natural number
- **e : t** ("e has type t")
 - "Hello" : string

$2 + 2 = 5$

3.5 is a natural number

$1 + 2$: string

We define what instances of a judgment hold
with *inference rules*

$$\frac{\text{if everything above this line is true}}{\text{then we can conclude everything below the line}} \begin{array}{l} \text{"premises"} \\ \text{(Rule Name)} \\ \text{"conclusion"} \end{array}$$

$$\frac{n \text{ is a natural number}}{n + 1 \text{ is a natural number}}$$

A rule with no premises is called an *axiom*:

$$\frac{}{0 \text{ is a natural number}}$$

We *derive* new facts by putting together rules from the system

$$\frac{}{0 \text{ is a natural number}} \qquad \frac{n \text{ is a natural number}}{n + 1 \text{ is a natural number}}$$

$$\frac{\frac{\frac{0 \text{ is a natural number}}{1 \text{ is a natural number}}}{2 \text{ is a natural number}}}{3 \text{ is a natural number}}$$

“derivation”

Another example

- **X** is XML

$$\frac{}{"" \text{ is XML}}^{(X-1)}$$

$$\frac{}{\langle T \rangle \text{ is XML}}^{(X-2)}$$

$$\frac{x \text{ is XML}}{\langle T \rangle X \langle /T \rangle \text{ is XML}}^{(X-3)}$$

$$\frac{x \text{ is XML} \quad y \text{ is XML}}{XY \text{ is XML}}^{(X-4)}$$

- is XML: $\langle \text{xml} \rangle \langle / \text{xml} \rangle$, $\langle \text{xml} \rangle \langle a \rangle \langle / a \rangle \langle b \rangle \langle c \rangle \langle / b \rangle \langle / \text{xml} \rangle$

- not XML: $\langle \text{xml} \rangle \langle / \text{html} \rangle$,
 $\langle a \rangle \text{Hi!} \langle / a \rangle$

$$\frac{\frac{\frac{}{\text{is XML}}^{(X-1)}}{\langle a \rangle \langle / a \rangle \text{ is XML}}^{(X-3)} \quad \frac{\frac{}{\langle c \rangle \text{ is XML}}^{(X-2)}}{\langle b \rangle \langle c \rangle \langle / b \rangle \text{ is XML}}^{(X-3)}}{\langle a \rangle \langle / a \rangle \langle b \rangle \langle c \rangle \langle / b \rangle \text{ is XML}}^{(X-4)} \quad \frac{}{\langle \text{xml} \rangle \langle a \rangle \langle / a \rangle \langle b \rangle \langle c \rangle \langle / b \rangle \langle / \text{xml} \rangle \text{ is XML}}^{(X-3)}$$

Inference rules can have other judgments as premises

$$\frac{n \text{ is a natural number}}{n \geq n} (\text{GE-EQ})$$

$$\frac{m+1 \text{ is a natural number} \quad m \geq n}{m+1 \geq n} (\text{GE-+})$$

$$\frac{\frac{\frac{\frac{0 \text{ is a nat}}{1 \text{ is a nat}}{2 \text{ is a nat}}{3 \text{ is a nat}}}{\frac{\frac{0 \text{ is a nat}}{1 \text{ is a nat}}{2 \text{ is a nat}}}{\frac{0 \text{ is a nat}}{1 \text{ is a nat}} \quad 0 \geq 0} (GE - EQ)} (GE +)} (GE +)} (GE +)$$

We can prove things about *all possible derivations* by induction

- If n is a natural number, $n \geq 0$.
 - How do we know?
 - If n is a natural number, then there is a derivation *saying* it's a natural number
- First, let's review induction

Induction on natural numbers (idea)

- A natural number is either zero or a natural number plus one
 - (this is true whether we use our inference rule definition of natural numbers or the normal math one)
- For a statement P :
 - Prove it's true for 0
 - Prove that if it's true for any n , it's true for $n + 1$
 - 
"Induction hypothesis"

Induction review

Theorem: For all $n \in \mathbb{N}$, $1 + 2 + \dots + 2^n = 2^{n+1} - 1$.

Proof:

Base case ($n = 0$): $1 = 2^{0+1} - 1 = 2^1 - 1 = 2 - 1 = 1$ ✓

Inductive case ($n + 1$):

Assume (Induction Hypothesis): $1 + 2 + \dots + 2^n = 2^{n+1} - 1$

Need to show: $1 + 2 + \dots + 2^n + 2^{n+1} = 2^{n+2} - 1$

$$\underbrace{1 + 2 + \dots + 2^n}_{2^{n+1} - 1} + 2^{n+1} = 2^{n+2} - 1$$

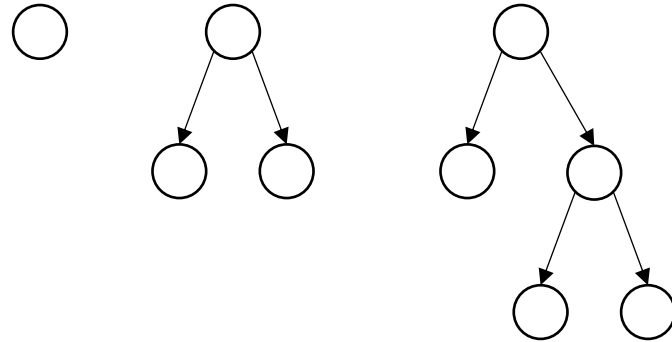
$$2 \cdot 2^{n+1} = 2^{n+2} \quad \checkmark$$



We can induct on *structures* (*structural induction*), not just numbers

Tree: Node with zero or more children, all of which are trees

Edge: Link between a node and its child (no other edges).



Theorem: The number of edges in a tree is one less than the number of nodes.

Put another way...

- A tree is either a leaf (no children) or a “root” node with one or more children, which are trees
- For a statement P:
 - Prove it’s true for a leaf
 - Prove that if it’s true for all of a root node’s children, it’s true for the tree



“Induction hypothesis”

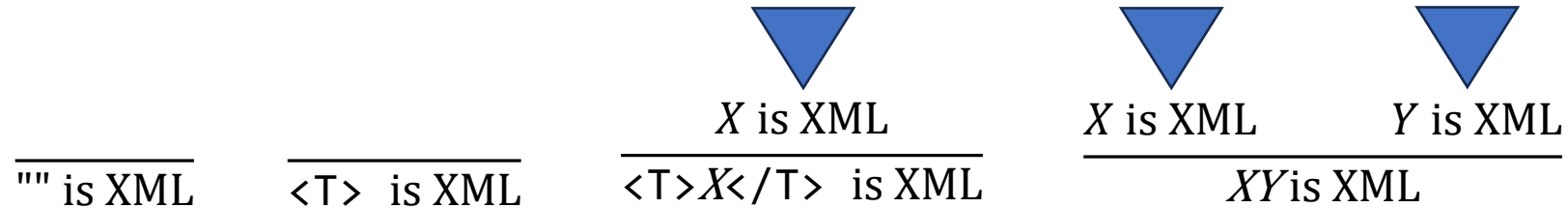
- Theorem: The number of edges in a tree is one less than the number of nodes.
- Proof:
 - Base case (Leaf): 1 node, 0 edges
 - Inductive case (Root w/ children):
 - Nodes: $1 + \text{nodes of child 1} + \dots + \text{nodes of child } n$
 - Edges: $n + \text{edges of child 1} + \dots + \text{edges of child } n$
 - $n + \text{edges of child 1} + \dots + \text{edges of child } n$
 - $= \cancel{n} + (\text{nodes of child 1} - \cancel{1}) + \dots + (\text{nodes of child } n - \cancel{1})$ by Induction Hypothesis (IH)
 - $= \text{nodes of child 1} + \dots + \text{nodes of child } n$
 - $= \text{Nodes} - 1$



Basic recipe for induction (structural or otherwise)

- Loosely: can do induction on any object that is built from smaller versions of itself as long as they eventually reach “0”
- Write a case for each way the object can be built
 - “Base case”: 0, objects not built from other objects
 - Just prove it directly
 - “Inductive case”: Object built from other objects ($n + 1$, root + children)
 - Assume that the statement is true for the smaller objects (IH)
 - Prove that it’s true for the bigger object
 - Usually using the IH... not always, but if you’re not using it, be very suspicious

Induction on derivations is called *rule induction*



A derivation is *either* an axiom (base case)
or an inference rule with one or more derivations (inductive case)

Induction on derivations is called *rule induction*

$$\frac{}{"" \text{ is XML}} \quad \frac{}{\langle T \rangle \text{ is XML}} \quad \frac{X \text{ is XML}}{\langle T \rangle X \langle /T \rangle \text{ is XML}} \quad \frac{X \text{ is XML} \quad Y \text{ is XML}}{XY \text{ is XML}}$$

Theorem: If X is XML, then X has the same number of open and close angle brackets.

Proof: By induction *on the derivation of* X is XML

You have a case for each way a derivation can be built (each inference rule)

Case $\frac{}{"" \text{ is XML}}$

Case $\frac{}{\langle T \rangle \text{ is XML}}$

Case $\frac{X \text{ is XML}}{\langle T \rangle X \langle /T \rangle \text{ is XML}}$

Case $\frac{X \text{ is XML} \quad Y \text{ is XML}}{XY \text{ is XML}}$

$$\frac{}{0 \text{ is a natural number}}(\text{N-0})$$

$$\frac{n \text{ is a natural number}}{n+1 \text{ is a natural number}}(\text{N-+})$$

$$\frac{n \text{ is a natural number}}{n \geq n}(\text{GE-EQ})$$

$$\frac{m+1 \text{ is a natural number} \quad m \geq n}{m+1 \geq n}(\text{GE-+})$$

Theorem: If n is a natural number, then $n \geq 0$

Proof (rule induction): By induction on the derivation of “ n is a natural number”